

Guide: Running and Verifying Mosquitto MQTT Broker on macOS

This document explains how to install, start, verify, and troubleshoot the Mosquitto MQTT broker on macOS for ESP32 or IoT projects.

1. Installing Mosquitto on macOS

You can install Mosquitto using Homebrew (recommended). If Homebrew is not installed, install it first from <https://brew.sh>.

Install Homebrew (if not installed)	<code>/bin/bash -c "\$(curl -fsSL https://raw.githubusercontent.com/Homebrew/homebrew-core/master/install.sh)"</code>
Install Mosquitto	<code>brew install mosquitto</code>
Start Mosquitto service	<code>brew services start mosquitto</code>

2. Running Mosquitto Manually (Debug Mode)

1. Open Terminal.
2. Start Mosquitto manually with verbose output:
`mosquitto -v`
3. Or specify configuration file explicitly:
`mosquitto -c /usr/local/etc/mosquitto/mosquitto.conf -v`
4. Press Ctrl + C to stop the broker when done.

3. Verifying Mosquitto is Running

Check service status	<code>brew services list</code>
Check listening ports	<code>sudo lsof -iTCP -sTCP:LISTEN -n -P grep 1883</code>
Test subscribe (new Terminal)	<code>mosquitto_sub -h localhost -t test/topic -v</code>
Test publish (another Terminal)	<code>mosquitto_pub -h localhost -t test/topic -m 'hello world'</code>

4. Allowing Mosquitto in macOS Firewall

1. Go to System Preferences → Security & Privacy → Firewall → Firewall Options.
2. If Firewall is on, click the '+' icon and add Mosquitto (usually at `/usr/local/sbin/mosquitto`).
3. Allow incoming connections for Mosquitto.
4. You may need to restart Mosquitto for the changes to take effect.

5. Troubleshooting

Mosquitto won't start	Check logs using 'brew services log mosquitto' or run 'mosquitto -v' m
ESP32 can't connect	Make sure Mac and ESP32 are on the same Wi-Fi network and firewa
Connection refused	Ensure 'listener 1883' and 'allow_anonymous true' are set in mosquitt
Permission denied	Run Mosquitto with sudo if port 1883 requires elevated privileges.

Tip: To automatically start Mosquitto on login, keep 'brew services start mosquitto' enabled. To stop it, run 'brew services stop mosquitto'.

© Drona Aviation | MQTT macOS Setup Guide